

DIY OBD1 ABA ECU remapping.

The first thing I have to mention is that all the info contained within this document is based on my own experiences. I am not a computer engineer, nor am I a member of Mensa. I am just a regular guy with the inability to leave well enough alone. I assume no responsibility for anything that may happen while you play with the code in your ECU. This includes blown engines ;). I'm hoping you've got some experience with standalone EFI tuning, because this document is written to help, not to teach. This document may not be 100% exact either, since there's not really any roadmap to doing this sort of thing, so always pay attention to how your car behaves. Now, on to the good stuff.

Motronic is a funny bird. Even the older stuff is kind of complex, and tuning it requires a basic knowledge of how it all works together. It's not like working with Megasquirt or any other standalone EFI, mainly because of the MAF. OBD1 management (referred to as M2.9 from this point on) uses three things to determine its fuel variable: throttle position, MAF/Load, and RPM. In the code there is a single 16x16 TPS-based fuel map, and many MAF/Load-based maps. The MAF/Load maps all work off the TPS map, so consider it a coarse scalar. If you want to go richer all over, raise the TPS map. Vice versa for leaner. The MAF/Load maps are where the fine tuning is done, and there are many of them to work with: idle fuel, part throttle, and WOT. They're pretty self explanatory ;). Idle is up to 14% throttle position (under load), part-throttle (the biggest of the MAF/Load maps) is from 15%-70%, and WOT is 70%-100% (more or less).

The MAF/Load fuel maps (idle, PT, WOT) should be touched if the coarse scalar didn't do the trick. Say you've turbocharged the car and now have different airflow requirements. DO NOT TUNE FOR BOOST WITH THE TPS TABLE! If you do make 'adjustments' that way, you'll be tuning for days to get everything right. For boost, you want to make sure you only add fuel when it's needed, and since boost can come on at part-throttle conditions, be sure to use the MAF/Load maps to handle any tuning required. You'll need to concentrate on the upper portions of the PT map (from '100' and up IMO), and all of the WOT map. The stock OBD1 ABA only goes as high as '150' in the WOT maps, so there is plenty of room to grow when adding boost. The important part here is to watch it trace in tunerpro (if you have an emulator), or to see where your AFR gauge tells you it's going lean. VAG-COM is a great tool to have for datalogging while tuning, but you can't get any O2 data (you unplugged the O2 sensor, remember ;)). The stock ECU stops recording O2 info at WOT anyway, so it's fairly useless to begin with. A wideband is HIGHLY recommended.

TunerPro RT - AE_NO_EGR.bin XDF File - OBD1 ABA GENERIC.xdf				
File XDF View Compare Acquisition Tools Window Help				
Parameter Summary List				
Item Name	Location	Hex	Calculated	Adjust
Warmup Enrichment #1	AE06->AE0B	(Table)	(Table)	
Warmup Enrichment #2	AE14->AE19	(Table)	(Table)	
Warmup Enrichment #3	AE22->AE27	(Table)	(Table)	
Warmup Enrichment #4	AE30->AE35	(Table)	(Table)	
Cranking Enrichment? #1	B6DD->B6E2	(Table)	(Table)	
Cranking Enrichment? #2	B6EB->B6F0	(Table)	(Table)	
Cranking Enrichment? #3	B6F9->B6FE	(Table)	(Table)	
Cranking Enrichment? #4	B707->B70C	(Table)	(Table)	
Mid/High throttle ignition #1	B8BD->B8BC	(Table)	(Table)	
Mid/High throttle ignition #2	B9E1->BAE0	(Table)	(Table)	
Mid/High throttle ignition #3	BB05->BC04	(Table)	(Table)	
Mid/High throttle ignition #4	BC29->BD28	(Table)	(Table)	
Idle/low throttle ignition #1	BD46->BDE1	(Table)	(Table)	
Idle/low throttle ignition #2	BDFF->BE9A	(Table)	(Table)	
Injector Latency.	AB79->AB7D	(Table)	(Table)	
Base/TPS fuel	AC98->AD97	(Table)	(Table)	
Mid/High throttle fuel #1	AEDC->AFDB	(Table)	(Table)	
Mid/High throttle fuel #2	B000->B0FF	(Table)	(Table)	
Mid/High throttle fuel #3	B124->B223	(Table)	(Table)	
Mid/High throttle fuel #4	B248->B347	(Table)	(Table)	
WOT Fuel	B36C->B46B	(Table)	(Table)	
Idle Fuel	AE98->AEB7	(Table)	(Table)	
Speed Limiter (?)	A456	B7	183	
Rev Limiter	A449	8F	6410	
Closed loop on/off (28=off/59=on)	6CB3	3B	59.00	
Idle speed based on coolant temp #1	C039->C03C	(Table)	(Table)	
Idle speed based on coolant temp #2	C059->C05C	(Table)	(Table)	
Idle speed based on batt volts.	C044->C048	(Table)	(Table)	
Acceleration pump/enrichment	ADA6->ADBD	(Table)	(Table)	
ABA M2.9	CF00-->CF01	(Checksum)	(Checksum)	

Not Emulating

Hardware: Flash & Burn v5.217.F

DA: Not Connected

16x16 Table, Offset: AC98, Cell Size: 8 Bit

Scaling for larger injectors:

In order to scale for larger injectors, consider lowering the TPS fuel table by “X” amount. I say “X” because it’s going to be a trial-and-error process here. Be sure to disable/unplug the O2 sensor before making any changes, and that the car is up to operating temp. Try to get the AFRs to 13.5-14.5:1 at idle, as this should allow the O2 enough leeway to compensate when you enable it/plug it back in. Most of the fuel tuning should be done with the O2 off anyway, so leave it off until you’re done IMO. Don’t worry if the numbers look too high or too low, rely on what the wideband is saying, and how the car runs. It’ll tell you when it’s happy.

There are ‘two’ (well actually 8, because each map is quadrupled for some reason) engine temp enrichment maps that need modifying as well. Again, these are a trial-and-error process. The car should fire up without these maps being touched, but the wideband will tell you that you’re insanely rich until it starts to get warm. Try changing all 8 by the same % to start. One of them is the warm up enrichment, and the other is cranking/initial enrichment. I’ve gotten my car to fire up and idle at 14:1 within a minute of startup, so it’s definitely possible to hit the right values with some experimentation.

Idle speed maps:

There are three idle speed maps. Two based on engine temp, and one based on voltage. Change all three when you want to effect idle speed changes. The two engine temp-based ones are what tells the ECU to raise the idle speed from a cold start, so you'll really only have to change the last two values in the map. The voltage based one seems to be a 'default' idle speed, or a low limit value. Whatever you set your car to idle at with the engine temp maps, set the voltage map to the same value.

Ignition maps:

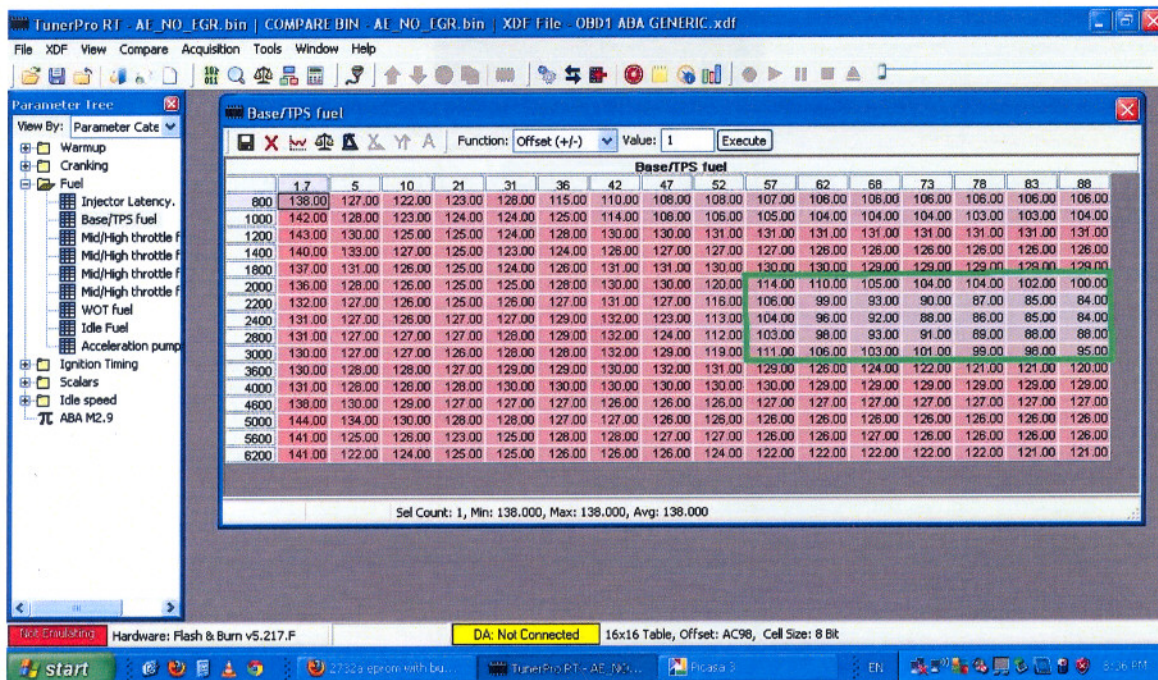
The only ignition map that should be touched (IMO) is the 'main' one. There is a low-part throttle map, but it BARELY uses it from what I can see. When driving, tunerpro only seems to trace on the main map anyway. The stock ABA ignition mapping is VERY conservative overall, since it's set up for 87 octane. All of the aftermarket chiptuners bump this map up a fair bit in their naturally aspirated chips, since they're set for 91 octane. As far as tuning for boost, in my experience I've left the stock ignition map in place and run 94 octane. The knock sensor doesn't seem to be very active up to 10psi with that octane/timing. You can tell when the ks is pulling timing by logging with VAG-COM. Watch the ignition plot, any sudden drops when the RPMs/Throttle are increasing or staying WOT indicate that the ECU is pulling timing to compensate for knock.

Injector Latency:

You shouldn't touch this map unless you're 100% certain you know what you're doing, as it affects the way the injectors function under different voltages. It can be used as a scalar for large injectors as well, but I HIGHLY recommend you don't do that.

Base/TPS fuel:

Ok, the good stuff. OBD1 uses many variables to arrive at the end fuel calculation, one of them being throttle position vs. RPM. If tuning for a naturally aspirated setup, you shouldn't have to touch this. There is a low spot in the map, near WOT, that actually leans out the fueling, so it might be a good idea to richen this up a bit (increase the values) if you're lean up top, instead of touching the airflow maps. It's 100% necessary (IMO) to bring this dip up to the level of the other cells around it if you're tuning for boost, otherwise you'll have nothing but headaches while tuning. This is the way I've tuned my own car, and it works great. See the picture below for clarification.



I am running 358cc injectors on my car, and I have subtracted 50 from all the cells, except for the dip, where I've put all the values to '80'. This seems to work great. This is just to get the car to run with the larger injectors though, the fine tuning is done on the Airflow/Load maps (idle, part throttle, WOT).

Airflow/Load Maps:

These maps are where the fine tuning will be done. They're actually 'load' based, and not MAF-based, but hey, how they work is the important part, not what they're named ;). They use TPS, airflow, and other things I'm sure, to come up with a 'Load' sum, which is what this map uses, as well as RPM. There is no real calculation to come up with a recognizable value for the axis, so I've left it in raw numbers. In stock form, at WOT, a bone-stock ABA will hit the '150' row at torque peak, and stay pretty close to that all the way to redline. The maps go to '220', so there's room to play when adding boost. Too much boost and you'll max out the map, and the ecu will be 'stuck' on the last column for it's fuel calculations, which is why OBD1 setups on a stock MAF can run VERY rich at WOT, especially if the turbo flows a lot of air, or the boost is cranked up. It's done that way on purpose, for safety. If you plan to max out the map, get a 3" MAF. I have no tips on tuning for that however, since I have not done it yet.

Part throttle boost can easily come on around the '100' column, and earlier, so be sure to address that when boost tuning as well. Make the same changes in the part-throttle and WOT maps.

Rev limit, Speed limiter, Closed loop on/off:

These three items are called 'scalars'. The rev limit can be raised to whatever you want, but keep in mind that the highest row on the maps is 6200 RPM. This means that if you're planning on spending a lot of time ABOVE that, because you've raised the limiter too high, then the ecu won't know exactly what to use for it's calculations. I can write files with a much higher RPM column on the maps, but it's a little too tough to explain how to do. Message me on whatever forum you've found this on, and I'll help out if needed. The speed limiter is set to 183km/h stock, so by moving it to 255km/h, you essentially delete it. The closed loop on/off is for tuning. **DO NOT DO ANY FUEL TUNING WITH THE O2 PLUGGED IN/TURNED ON.** The ecu will try to compensate and you'll get many CELs. This bit of code allows you to turn the O2 data 'off' without unplugging the sensor. When you're done with the tuning, simply turn it back 'on'.

Summary:

So, that's about all you'll need to know to get started, outside of the actual hardware useage/chip burning. It's hard to go into much detail in that regard, because there are so many ways of doing it. I prefer to live tune with the Moates.net 'Ostrich 2.0' emulator, which is only \$170 (cheapest ABA tuning available ;)), but it's also possible to tune with just a chip burner. It'll take longer, but it's possible.

Tuning Summary:

For the tl;dr crowd ;).

Tuning NA:

1. Adjust TPS/Base fuel table for injectors (if larger than stock).
2. Adjust warmup/cranking if larger injectors used.
3. Add timing to main timing map at WOT and cruise areas.
4. Enjoy.

Tuning for boost:

1. Adjust TPS/Base fuel table for larger injectors and remove 'dip' at end of table.
2. Adjust warmup/cranking enrichment.
3. Adjust Main Timing map according to compression ratio/gas/boost
4. Adjust fuel on part-throttle and WOT maps to give proper AFRs under boost.
5. Enjoy.